Abnormal molecular mass

1. The Van't Hoff factor will be highest for

(a) Sodium chloride

(b) Magnesium chloride

(c) Sodium phosphate

(d) Urea

2. Which of the following salt has the same value of Van't Hoff factor i as that of $K_3[Fe(CN)_6]$

(a) $Al_2(SO_4)_3$ (b) $NaCl$

(c) Na_2SO_4 (d) $Al(NO_3)_3$

3. When benzoic acid dissolve in benzene, the observed molecular mass is

(a) 244 (b) 61

(c) 366 (d) 122

4. The ratio of the value of any colligative property for KCl solution to that for sugar solution is nearly

(a) 1 (b) 0.5

(c) 2.0 (d) 3

5. Van't Hoff factor of $Ca(NO_3)_2$ is

(a) 1 (b) 2

(c) 3 (d) 4

6. Dry air was passed successively through a solution of $5gm$ of a solute in $80gm$ of water and then through pure water. The loss in weight of solution was $2.50gm$ and that of pure solvent $0.04gm$. What is the molecular weight of the solute

(a) 70.31 (b) 7.143

(c) 714.3 (d) 80

7. The Van't Hoff factor calculated from association data is always...than calculated from dissociation data

(a) Less

(b) More

(c) Same

(d) More or less

8. If α is the degree of dissociation of Na_2SO_4, the Vant Hoff's factor (i) used for calculating the molecular mass is

(a) $1+\alpha$ (b)

(c) $1+2\alpha$ (d) $1-2\alpha$

9. Van't Hoff factor i

(a) $= \frac{\text{Normal molecular mass}}{\text{Observed molecular mass}}$

(b) $= \frac{\text{Observed molecular mass}}{\text{Normal molecular mass}}$

(c) Less than one in case of dissociation

(d) More than one in case of association

10. Which of the following compounds corresponds Van't Hoff factor $'i'$ to be equal to 2 for dilute solution
(a) K_2SO_4 (b) $NaHSO_4$
(c) Sugar (d) $MgSO_4$

11. The Van't Hoff factor i for a 0.2 molal aqueous solution of urea is
(a) 0.2 (b) 0.1
(c) 1.2 (d) 1.0

12. One mole of a solute A is dissolved in a given volume of a solvent. The association of the solute take place according to $nA \rightleftarrows (A)_n$. The Van't Hoff factor i is expressed as
(a) $i = 1 - x$ (b) $i = 1 + \frac{x}{n}$
(c) $i = \frac{1-x+\frac{x}{n}}{1}$ (d) $i = 1$

13. Acetic acid dissolved in benzene shows a molecular weight of
(a) 60 (b) 120
(c) 180 (d) 240

14. The observed osmotic pressure of a solution of benzoic acid in benzene is less than its expected value because
(a) Benzene is a non-polar solvent
(b) Benzoic acid molecules are associated in benzene
(c) Benzoic acid molecules are dissociated in benzene
(d) Benzoic acid is an organic compound

15. The experimental molecular weight of an electrolyte will always be less than its calculated value because the value of Van't Hoff factor "i" is
(a) Less than 1
(b) Greater than 1
(c) Equivalent to one
(d) Zero

16. The molecular mass of acetic acid dissolved in water is 60 and when dissolved in benzene it is 120.This difference in behaviour of CH_3COOH is because
(a) Water prevents association of acetic acid
(b) Acetic acid does not fully dissolve in water
(c) Acetic acid fully dissolves in benzene
(d) Acetic acid does not ionize in benzene

17. The correct relationship between the boiling points of very dilute solutions of $AlCl_3(t_1)$ and $CaCl_2(t_2)$, having the same molar concentration is
(a) $t_1 = t_2$ (b) $t_1 > t_2$

(c) $t_2 > t_1$ (d) $t_2 \geq t_1$

18. The Van't Hoff factor for sodium phosphate would be

(a) 1 (b) 2

(c) 3 (d) 4

19. The molecular weight of benzoic acid in benzene as determined by depression in freezing point method corresponds to

(a) Ionization of benzoic acid

(b) Dimerization of benzoic acid

(c) Trimerization of benzoic acid

(d) Solvation of benzoic acid